

# Mohammad Taufeeque

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## EDUCATION

### Indian Institute of Technology Bombay

2018–2022

B.Tech (with Honors) in Computer Science and Engineering

Mumbai, India

– GPA - **9.24**/10.0

– Bachelor's Thesis - Fianchetto: Speed, Belief, Guile, Caution to Win at Reconnaissance Blind Chess

## SELECTED PUBLICATIONS

- [1] **M. Taufeeque**, S. Heimersheim, A. Gleave, and C. Cundy, *The Obfuscation Atlas: Mapping Where Honesty Emerges in RLVR with Deception Probes*, in **ICML 2026, Spotlight**
- [2] **M. Taufeeque**, A. D. Tucker, A. Gleave, and A. Garriga-Alonso, *Path Channels and Plan Extension Kernels: a Mechanistic Description of Planning in a Sokoban RNN*, in **ICLR 2026, Spotlight at Mechanistic Interpretability Workshop, NeurIPS 2025**
- [3] A. Tamkin, **M. Taufeeque**, and N. Goodman, *Codebook Features: Sparse and Discrete Interpretability for Neural Networks*, in **ICML 2024**
- [4] **M. Taufeeque**, P. Quirke, M. Li, C. Cundy, A. D. Tucker, A. Gleave, and A. Garriga-Alonso, *Planning in a recurrent neural network that plays Sokoban*, **Mechanistic Interpretability Workshop, ICML 2024**
- [5] K. Pelrine\*, **M. Taufeeque\***, M. Zając, E. McLean, and A. Gleave, *Exploiting Novel GPT-4 APIs*. arXiv: 2312.14302
- [6] A. Gleave\*, **M. Taufeeque\***, J. Rocamonde\*, E. Jenner, S. H. Wang, S. Toyer, M. Ernestus, N. Belrose, S. Emmons, and S. Russell, *imitation: Clean Imitation Learning Implementations*. arXiv: 2211.11972
- [7] G. Perrotta, R. W. Gardner, C. Lowman, **M. Taufeeque**, N. Tongia, S. Kalyanakrishnan, et al., *The Second NeurIPS Tournament of Reconnaissance Blind Chess*, in **NeurIPS 2021 Competitions Track**
- [8] **M. Taufeeque\***, S. Koita\*, N. Spicher, and T. M. Deserno, *Multi-camera, multi-person, and real-time fall detection using long short term memory*, in **SPIE Medical Imaging 2021**

## INDUSTRY & RESEARCH EXPERIENCE

### FAR.AI - Research Engineer

Berkeley, USA

Supervisor: Dr. Adam Gleave

Aug '22 - Present

- Implemented **RLVR with probe-based reward** & managed **1000+** experiments with **10,000+ GPU hours**
- Reverse-engineered the **planning algorithm** learned by an RNN to play Sokoban through model-free RL
- **Codebook Features**: Developed an interpretability tool that discretizes NN activations into features
- Red-teaming: Found universal jailbreaks for **GPT-5** & **Claude Opus 4**, and novel exploits for **GPT-4** APIs
- Co-developed & benchmarked *imitation*, an open-source imitation learning library with **1.7k+ GitHub stars**

### Microsoft Research - Research Intern

Bangalore, India

Guide: Prof. Sunita Sarawagi (IIT Bombay) & Dr. Sriram Rajmani (Microsoft Research)

Dec '21 - Aug '22

- Developed an online algorithm against OOD data to integrate noisy feedback rules to a trained ML text classifier
- Deployed the algorithm to **Microsoft's Ads** system & improved the compliance of the system by over **55%**

### Goldman Sachs - Summer Analyst

Bangalore, India

Guide: Kesavan Mukunthan

Summer 2021

- Built a webapp to compute & display **factor-based performance exposures** for mutual fund portfolios

- Developed an application that **detects falls in real-time** using human pose keypoints from multiple cameras
- Maintainer of the open-source project on GitHub with **325+ stars & 68+ forks**

## AI Agent for Reconnaissance Blind Chess

IIT Bombay

Bachelor's Thesis | Guide: Prof. Shivaram Kalyanakrishnan | Won NeurIPS competition (RBC) Aug '21 - Dec '21

- Developed an AI Agent for RBC, a variant of Chess where only a 3x3 region can be sensed before making a move
- Won the **NeurIPS 2021** competition on RBC with **91.3%** win rate & **100 Elo pts** margin from the runner-up

## Randomized Planning Algorithms for POMDPs

IIT Bombay

Guide: Prof. Shivaram Kalyanakrishnan

Spring 2021

- Designed planning algorithms for **POMDPs** that achieved **20%** higher rewards than the SoTA algorithms

## SAFE App Vulnerabilities

IIT Bombay

Guide: Prof. Bhaskaran Raman

Aug '20 - Apr '21

- Found **severe vulnerabilities** in the SAFE App IITB, used by many institutions to conduct remote exams
- Reported **data-leak**, **APK signature verification** and **timing-based** vulnerabilities in the Android app

## SCHOLASTIC ACHIEVEMENTS

|                                   |                                                       |      |
|-----------------------------------|-------------------------------------------------------|------|
| Attended Google Research Week     | Selected among 50 undergraduates nationwide           | 2022 |
| Best AI Agent out of 18 Bots      | RBC chess competition in NeurIPS 2021                 | 2021 |
| AP (Advanced Performer) Grade     | Best performance in Machine Learning course (GNR 638) | 2020 |
| All India Rank 303                | JEE Advanced (230,000 aspirants)                      | 2018 |
| All India Rank 330                | JEE Mains (1.2 Million aspirants)                     | 2018 |
| Merit-cum-Means (MCM) Scholarship | IIT Bombay                                            | 2018 |
| National top 1%                   | Indian National Physics Olympiad (INPhO)              | 2017 |
| National top 1%                   | Indian National Chemistry Olympiad (INChO)            | 2017 |
| KVPY Science Fellowship           | Government of India                                   | 2016 |

## TECHNICAL SKILLS

**ML & Infra** PyTorch, JAX, Transformers, TransformerLens, Accelerate, PEFT, WandB, Kubernetes

## MENTORSHIP & TEACHING

### Research Manager at Impact Academy

Apr '24 - Apr '25

- Provided technical guidance & managed research projects at this AI Safety field-building org

### Teaching Assistantships

- Intro to ML Safety (Summer 2022 & Spring 2023)
- Artificial Intelligence & Machine Learning (Autumn 2021)
- Medical Image Computing (Spring 2022)
- Calculus II (Autumn 2021)

### English Language Tutor at ELIT IIT Bombay

Autumn 2020

- Designed an English language curriculum and held classes for 50 undergraduates registered for the program

### Member in Developer's Community (DevCom) IIT Bombay

Jan '19 - Aug '20

- Maintained, developed & updated features of **InstiApp**, the institute app with **10,000+** downloads

### Mentor of SoC project - Intrusion Detection System | Github: taufeeque9/IDS

Summer 2020

- Mentored a team of **9 developers** in building a real-time system to monitor networks for malicious activity